# UNIT CIRCLE VOLVELLE & TRIGONOMETRIC CALCULATOR

Special Right Triangles, Radians, Coordinates & Wave Projections

SERIES 1 • CARD 08 / 12 • DOMAIN: UNIT CIRCLE TRIGONOMETRY & PROJECTIONS

CARD 08 / 12 • VOLV-01

MATHEMATICAL GUIDE

## 1. CONCEPT & GEOMETRIC PROOF

**CIRCULAR COORDINATES & DYNAMIC PROJECTIONS**

Trigonometry is circular motion, not triangles.

On unit circle (r = 1), coordinates are trig values:

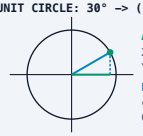
**(X, Y) = (cos θ, sin θ)**

Rotating disc projects angle θ onto cos/sin!

**CALIBRATED 84.00 MM RADIUS DATUM**

- Outer Ring: 84.00 mm ⌀ (Matches Fraction 1 Whole)
- Dual Scales: Degrees (0°–360°) and Radians (0–2π)
- Special Angles: Exact cutouts for 30°, 45°, and 60°

**UNIT CIRCLE: 30° –> (cos = √3/2 ≈ 0.866, sin**



**ANGLE θ = 30° (π/6)**

X = cos(30°) = √3/2 ≈ 0.8

Y = sin(30°) = 1/2 = 0.50

**Pythagorean Identity:**

cos²(θ) + sin²(θ) = 1

0.75 + 0.25 = 1.000

## 2. KIT PARTS INVENTORY (6 PIECES)

| COMPONENT | QTY | SPEC | COLOR |
|---|---|---|---|
| Unit Circle Disc | 1x | 84.0 mm ⌀ | PLA White |
| Radian Overlay | 1x | Radial Cursor | Clear PETG |
| Projection Arm | 1x | Windows | Clear PC |
| Center Pivot | 1x | Brass Bushing | Solid Brass |
| Stand & Lid | 1x | Desk Stand | Matte Charcoal |
| 3x1 Storage Bin | 1x | 49.0 mm | Matte Charcoal |

**TOTAL: 4 Volvelle Components + Storage Bin + Stand Lid = 6 Parts**

**QUADRANT SIGN CONVENTION**

- Quadrant I (+, +): All functions positive (0° to 90°).
- Quadrant II (-, +): Sine positive, Cosine negative (90°–180°).
- Quadrant III (-, -): Tangent positive, Sine/Cos negative.
- Quadrant IV (+, -): Cosine positive, Sine negative.

## 3. PACK-UP & SUPPORT

**✓ 10-SECOND PACK-UP**

Align transparent cursor to 0° / 360°, slip the volvelle into its protective sleeve, and lay flat into the box recess. Snap the stand lid shut for silent, rattle-free carry.

**✓ NO LOST PIECE GUARANTEE**

Classrooms lose pieces. Every single math component is permanently stocked for single replacements.

stackcalc.io/parts

Order replacements anytime

**✓ HANDS-ON TUTORIAL**

Flip over for 5-step concrete lesson & live 3D Video QR!

**Keep stored inside box lid!**

**FUNDAMENTAL THEOREM**

For any real angle θ on the unit circle: cos²(θ) + sin²(θ) = 1 identically.

**STACKCALC ARCHIVAL INSTRUMENT LABS**

Designed in Mountain View, California • Non-Toxic BPA-Free • Classroom Lab Stations: stackcalc.io/schools

**OFFICIAL REFILL PORTAL**

stackcalc.io/parts

# LESSON GUIDE: UNIT CIRCLE TRIGONOMETRY & PROJECTIONS

Step-by-Step Hands-On Tutorial: Angle 30° -> (cos 30° = √3/2, sin 30° = 1/2)

SERIES 1 • CARD 08 OF 12 • DOMAIN: UNIT CIRCLE TRIGONOMETRY & PROJECTIONS

CARD 08 / 12 • VOLV-01

LESSON TUTORIAL

**SCAN FOR 3D VIDEO DEMO**

OPEN CAMERA & SCAN

stackcalc.io/products/unit-circle-volvelle/

Zero app install • iPad & Chromebook

**CURRICULAR ALIGNMENT**

- CCSS HSF.TF.A.2 (Unit Circle)
- CCSS HSF.TF.A.3 (Special Tria...)
- TEKS 111.42.c.2.P (Trig Functio...)

Grades 9–12 Trigonometry & Physics

**GUIDING INQUIRY**

When the cursor rotates past 90° into Quadrant II, why does the horizontal X coordinate become negative while Y stays positive?

**1 STEP 1: ROTATE CURSOR TO 30° (π/6 RAD)**

Turn the transparent cursor arm to the 30° mark.

Look at the inner radian scale: 30° aligns exactly with π/6 rad.

Notice the right triangle projected onto the unit circle base!

**2 STEP 2: READ HORIZONTAL COSINE PROJECTION**

Look down at the horizontal X-axis window:

X = cos(30°) = √3 / 2 ≈ 0.8660.

This is the horizontal base length of the 30°-60°-90° triangle.

**3-4 STEPS 3 & 4: READ SINE & VERIFY ON STACKCALC**

**3. READ VERTICAL SINE: Look at vertical Y-axis window: Y = sin(30°) = 1/2 = 0.5000.**

**4. VERIFY PYTHAGOREAN IDENTITY: cos²(30°) + sin²(30°) = 0.75 + 0.25 = 1.0000!**

**Calculator Verification: Press Unit Circle Radian Icon ([COS], [SIN] & [⁻¹][•,x]) and run 45 [COS] [⁻¹] [x²] 45 [SIN] [⁻¹] [x²] [+] → 1.**

**5 CORE PRINCIPLE: SINE AND COSINE ARE CIRCLE COORDINATES**

Textbook right triangles only work for angles < 90°. The volvelle proves that sine and cosine are the literal (X, Y) coordinates of a point moving around a circle! In Quadrant II, cosine naturally becomes negative because X is negative, while sine remains positive!

**★ EXTENSION INVESTIGATION: SPECIAL ANGLE: 60° (π/3 RAD)**

Rotate cursor to 60°. Notice X and Y coordinates swap: X = cos(60°) = 1/2 = 0.500, Y = sin(60°) = √3/2 ≈ 0.866! (Scan QR for 3D animated solution!)

→ Assemble with physical blocks, then scan QR code for 3D animated solution.

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Interactive Mathematics Manipulative Ecosystem • Mountain View, CA • BPA-Free

**PARTS & REFILL PORTAL**

stackcalc.io/parts

# STACKCALC STEM PEDAGOGY & PHYSICAL COMPUTING

CONCRETE • REPRESENTATIONAL • ABSTRACT

### 1. TACTILE CONCRETE

**7.00mm 1U PITCH**

Every physical block dimension is locked to an exact integer ratio, providing tactile feedback of quantity.

**MOTOR KINESTHETIC**

Sliding, stacking, and interlocking builds spatial memory and intuitive number sense before symbols appear.

**CONSERVATION OF QUANTITY**

Mass, area, and volume remain invariant regardless of how blocks are grouped or rotated.

**SELF-CORRECTING**

Precision mechanical detents seat flush only when arithmetic identities and geometric proofs are correct.

**CCSS.MATH.PRACTICE.MP5**

Use appropriate tools strategically.

### 2. HARDWARE RPN STACK

**POSTFIX ARITHMETIC**

[ARG 1] [ENTER] [ARG 2] [OP]. Zero operator ambiguity, zero parentheses, and zero hidden precedence rules.

**X, Y, Z, T REGISTERS**

Every intermediate result is visible on the display stack. What is stacked on the desk is preserved in memory.

**DIGITAL TWIN OF TOWERS**

Physically stacking blocks onto a stage lid directly mirrors pushing values into the digital stack registers.

**STEP-BY-STEP REASONING**

Students observe each operator execute in real time, making compound calculations completely transparent.

**CCSS.MATH.PRACTICE.MP1**

Make sense of problems & persevere.

### 3. LEARNING CONTINUUM

**GRADES K–2: FOUNDATION**

Ten-Frames, Number Rack & Dominoes for subitizing, counting, 5-and-10 anchors, and place value fluency.

**GRADES 3–5: FRACTIONS**

Fraction bars, prime factorization towers, time gears, and discrete combinatorics with probability dice.

**GRADES 6–12: ADVANCED**

Algebraic factoring rails, unit circle volvelles, complex numbers, and professional scientific RPN workflows.

**DIFFERENTIATED TIERS**

Tactile physical manipulates bridge English language learners, visual learners, and accelerated STEM students.

**CCSS.MATH.PRACTICE.MP4**

Model with mathematics.

### CLASSROOM CARE & SANITIZATION

**1. FOOD-SAFE RECYCLED POLYMERS:**

Manufactured from high-purity virgin polymer (PLA / PETG). Non-toxic, odorless, BPA-free, and compliant with RoHS, EN71-3, and ASTM F963 safety standards.

**2. CLASSROOM CLEANING & DISINFECTION:**

Wash blocks in warm water with mild liquid dish soap (up to 55°C / 130°F) or wipe with standard classroom disinfectant wipes. Allow to air dry flat before storage.

**3. MODULAR STORAGE DETENTS:**

Stackable 3x1 and 3x3 stage lids feature multi-axis detents tested to 10,000+ opening cycles. Trays pack rattle-free for classroom mobile cart transport.

**4. SOCRATIC CLASSROOM DISCUSSION:**

Encourage students to physically construct models to test hypotheses before executing calculator calculations: "What invariant property did your physical model prove?"

# VOLV-01: IN-BOX SPECIFICATION & QC AUDIT

CALIFORNIA LAB QC

### CALIFORNIA LAB QC AUDIT CHECKLIST

**[✓] MITUTOYO DIGITAL CALIPER DATUM:**

All block dimensions conform to ±0.05 mm tolerance across universal 7.00 mm 1U visible height system.

**[✓] 1.85 MM RETENTION SOCKET INTERFERENCE:**

Stacking plugs seat flush with 0.15 mm diametral interference for zero-wobble tower verticality.

**[✓] ZERO-RATTLE DETENT LOCK:**

Multi-axis lid detents lock securely over 3x1 bin perimeter rim. Audible snap with 1.4 N retention.

**[✓] CONTINUOUS 360° MICRO-FLUTING & SEAM TRAP:**

2.0 mm pitch, 0.50 mm flutes. Slicer seam locked into rear-left corner seam trap notch.

**[✓] FOOD-SAFE RECYCLED POLYMER:**

100% virgin PLA / PETG. RoHS and EN71-3 compliant, non-toxic, odorless, child-safe.

**[✓] IN-BOX COLLATERAL PLACEMENT:**

Place this reference card inside the box lid or inside the storage tray.

**RPN HARDWARE COMPATIBILITY**

- Compatible with StackCalc SC-6 (Early STEM)
- Compatible with StackCalc SC-32 (Scientific RPN)
- Zero parentheses, zero operator precedence errors
- Physical towers are the exact digital twin of the stack!

**TOTAL KIT INVENTORY: 6 PIECES**

Verified by Assembly Tech & Machine Vision at Packing Bench.

**Inspection Pass • Mountain View, CA**

**LIFETIME REPLACEMENT GUARANTEE**

**NO LOST PIECE LEFT BEHIND™**

Classrooms lose blocks. We understand. Every individual manipulative block has a permanent part number. Visit stackcalc.io/parts to order exact replacement pieces with same-day dispatch ($1.99–$2.99/piece).

**PORTAL: stackcalc.io/parts**

**DISTRICT & CLASSROOM PROCUREMENT**

- Net-30 Invoicing for US Public & Private School Districts.
- Volume Classroom Bundles (10-packs & 30-packs).
- Free Teacher Activity Books & Lesson Blueprints.
- Title I STEM & Carl D. Perkins Grant eligible.

Email: procurement@stackcalc.io

stackcalc.io/schools

**3D INTERACTIVE VIDEO DEMOS**

Scan the QR code on the back of this card (Side B) to access:

- 60fps 3D animated walkthrough of each lesson
- Interactive web 3D model with explode view
- Step-by-step calculator keystroke simulations
- Printable student worksheets and answer keys

**VIDEOS: stackcalc.io/products**

Designed & Engineered in Mountain View, California